

Convexity Interpolation

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1 Theory

Let $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$. We say φ is convex at x if there exists a neighborhood U of x such that for all $z \in U$,

$$\varphi(z) - \varphi(x) \geq \nabla \varphi(x) \cdot (z - x)$$

The problem we are trying to solve is the following: Let $F_1, \dots, F_n : \mathbb{R} \rightarrow \mathbb{R}$ be convex functions and $r_1, \dots, r_n : \mathbb{R}^d \rightarrow \mathbb{R}$ be defined by $r_i = \|x - x_0^i\|$. We want to determine the maximum β_n such that

$$\frac{1}{2} \|x\|^2 + \sum_1^{n-1} \beta_i F_i \circ r_i + \beta_n F_n \circ r_n$$

is convex. Therefore, for all x , we need to find a neighborhood U_x such that for all $z \in U_x$,

$$\begin{aligned} \beta_n &\geq - \frac{(1/2) \|z\|^2 + \sum_1^{n-1} \beta_i F_i(r_i(z)) - (1/2) \|x\|^2 + \sum_1^{n-1} \beta_i F_i(r_i(x)) - \nabla((1/2) \|\cdot\|^2 + \sum_1^{n-1} \beta_i F_i \circ r_i) |_x \cdot (z - x)}{F_n(r_n(z)) - F_n(r_n(x)) - \nabla(F_n \circ r_n) |_x \cdot (z - x)} \\ &= - \frac{(1/2) \|z - x\|^2 + \sum_1^{n-1} \beta_i F_i(r_i(z)) - \beta_i F_i(r_i(x)) - \beta_i (F'_i(r_i)/r_i)(x - x_0^i) \cdot (z - x)}{F_n \circ r_n(z) - F_n \circ r_n(x) - (F'_n(r_n)/r_n)(x - x_0^n) \cdot (z - x)} \\ &= - \frac{(1/2) \|z - x\|^2 + \sum_1^{n-1} \beta_i F_i(r_i(z)) - \beta_i F_i(r_i(x)) - \beta_i f_i(r_i)(x - x_0^i) \cdot (z - x)}{F_n \circ r_n(z) - F_n \circ r_n(x) - f_n(r_n)(x - x_0^n) \cdot (z - x)} \end{aligned}$$

If the demonenator is zero, we define the fraction to be $-\infty$. Since z is close to x and hence $r_i(z)$ is close to $r_i(x)$, we can use the Taylor approximation. For $i \in [n]$, can approximate

$$F_i(r_i(z)) - F_i(r_i(x)) \approx F'_i(r_i(x))(r_i(z) - r_i(x)) + \frac{1}{2} F''_i(r_i(x))(r_i(z) - r_i(x))^2$$

and also

$$r_i(z) - r_i(x) \approx \nabla r_i |_x \cdot (z - x) = \frac{(x - x_0^i) \cdot (z - x)}{r_i(x)}$$

The terms cancel and we have

$$F_i(r_i(z)) - F_i(r_i(x)) - \nabla(F_i \circ r_i) |_x \cdot (z - x) \approx \frac{1}{2} F''_i(r_i(x))(r_i(z) - r_i(x))^2$$

Therefore, the fraction is approximately

$$\beta_n \geq - \frac{\|z - x_0^n\|^2 + \sum_1^{n-1} \beta_i F''_i(r_i(x))(r_i(z) - r_i(x))^2}{F''_n(r_n(x))(r_n(z) - r_n(x))^2}$$

If $x \neq x_0^n$, we can choose z in the neighborhood such that $\|z - x_0^n\| = r_n(z) = r_n(x) = \|x - x_0^n\|$. Hence, the denominator is zero and the fraction is $-\infty$. Therefore, to compute the maximum, we only need to consider $x = x_0^n$. We have

$$\beta_n \geq - \frac{\|z - x_0^n\|^2 + \sum_1^{n-1} \beta_i F''_i(r_i(x_0^n))(r_i(z) - r_i(x_0^n))^2}{F''_n(r_n(z))r_n(z)^2}$$

We can use the Taylor approximation

$$r_i(z) - r_i(x_0^n) \approx \frac{(x_0^n - x_0^i) \cdot (z - x_0^n)}{r_i(x_0^n)}$$

and hence

$$\beta_n \geq - \frac{\|z - x_0^n\|^2 + \sum_1^{n-1} \beta_i F''_i(r_i(x_0^n)) \left(\frac{(x_0^n - x_0^i) \cdot (z - x_0^n)}{r_i(x_0^n)} \right)^2}{F''_n(r_n(z))r_n(z)^2}$$

We want to maximize this fraction. Since $r_n(z) = \|z - x_0^n\|$, we can write this fraction as

$$- \frac{1}{F''(r_n(z))} - \frac{1}{F''(r_n(z))} \left(\sum_1^{n-1} \beta_i F''(r_i(x_0^n)) \left(\frac{(x_0^n - x_0^i) \cdot (z - x_0^n)}{\|x_0^n - x_0^i\| \|z - x_0^n\|} \right)^2 \right)$$

Since $r_n(z) \rightarrow 0$, we can approximate $F''(r_n(z))$ by $\lim_{r \rightarrow 0} F''(r)$. We can treat this value as a fixed constant. If F is defined by $F(z) = z \operatorname{erf}(z/\alpha) + (\alpha/\sqrt{\pi}) \exp(-z^2/\alpha^2)$ then $\lim_{r \rightarrow 0} F''(r) = 2/(\alpha\sqrt{\pi})$. Therefore, we only need to minimize the following quantity

$$\sum_1^{n-1} \beta_i F''(r_i(x_0^n)) \left(\frac{(x_0^n - x_0^i) \cdot (z - x_0^n)}{\|x_0^n - x_0^i\| \|z - x_0^n\|} \right)^2$$

We will make the following substitution for convenience. Let

$$c_i = \beta_i F''(r_i(x_0^n)) \quad x_i = \frac{x_0^n - x_0^i}{\|x_0^n - x_0^i\|} \quad u = \frac{z - x_0^n}{\|z - x_0^n\|}$$

The problem become

$$\min_u \sum_1^{n-1} c_i (x_i \cdot u)^2$$

subject to $\|u\| = 1$. We know $(x_i \cdot u)^2 = u^T x_i x_i^T u$, so we need to minimize

$$\min_{\|u\|=1} u^T \left(\sum_1^{n-1} c_i x_i x_i^T \right) u$$

This is a Rayleigh quotient, and the minimum is the smallest eigenvalue of

$$\sum_1^{n-1} c_i x_i x_i^T$$

(this is a symmetric real matrix so the smallest eigenvalue is guaranteed to exist).

2 Test

For simplicity, assume

$$\begin{aligned} F(z) &= z \operatorname{erf}(z) + (1/\sqrt{\pi}) \exp(-z^2) \\ d &= 1 \\ n &= 2 \\ \beta_1 &= 1 \\ x_0^1 &= -1 \\ x_0^2 &= 1 \end{aligned}$$

Therefore, we want to find the minimum β_2 such that

$$\frac{1}{2}x^2 + |x+1| \operatorname{erf}(|x+1|) + (1/\sqrt{\pi}) \exp(-(x+1)^2) + \beta_2 |x-1| \operatorname{erf}(|x-1|) + (1/\sqrt{\pi}) \exp(-(x-1)^2)$$

is a convex function. By calculation, we know $x_i = 1$ and $c_i = (2/\sqrt{\pi})e^{-4}$, so the minimum eigenvalue of $\sum c_i x_i x_i^T$ is simply c_i . Moreover, we know $\lim_{r \rightarrow 0} F''(r) = 2/\sqrt{\pi}$, so the beta threshold we calculated is

$$-\frac{1}{2/\sqrt{\pi}} (1 + (2/\sqrt{\pi})e^{-4}) = -0.905$$

I also plotted the function on desmos and the beta threshold it calculated is between -0.903 and -0.904 . The results are very close to each other. It also shows that F'' achieves its minimum at $x_0^2 = 1$, so this is the point closest to non-convexity.

